

B.Sc. Professional Project (COM3001)

Module Handbook for Students

Department of Computer Science

Sept. 2024

Document History		
Author	Date	Notes
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T. Giannetsos	3/10/2016	Revised
T. Giannetsos	17/09/2017	Revised
T. Giannetsos	16/09/2018	Revised
B. Dongol	11/10/2019	Revised
B. Dongol	24/11/2021	Revised
B. Dongol	20/09/2022	Revised
Guerin	24/09/2023	Revised
Guerin	21/09/2024	Revised

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Chapter 1

Introduction

This document is a point of reference for details related to the Professional Project module for current students. Any questions related to this document should be emailed to the Undergraduate Projects Coordinator (Frank Guerin, f.guerin@surrey.ac.uk).

It is designed to help you initially choose a project. During the academic year, it also helps you keep track of deliverables; a complete description of each deliverable is provided. Finally, this document specifies how the project is assessed, including pointers to marking criteria, details on the examination, as well as the process for dealing with complaints.

1.1 What does a project involve?

The Professional (or Final Year) Project is a module that allows you to undertake a major piece of work that will involve developing and applying material encountered on the course under the guidance of a supervisor, who is normally a member of the teaching staff. The Project lasts for the whole academic year and involves practical work, writing a report on the work done, and preparing a video presentation. The assessment is based on the report and presentation of the work reported therein.

This module is different from all other modules in that the content is determined to a large extent by you, the student. The Project gives you a lot of freedom in choosing what to study, but on the other hand it requires a lot more independent thought and organisational skills than the majority of modules. Therefore, the project should not be considered an easy component as most students find it more demanding than the usual lecture courses. However, it is also more rewarding and a well-executed project can give confidence and pride in the results. It is also something that can be used to demonstrate ability to potential employers.

1.2 Aim and Purpose of Project

The aim of the Project is to demonstrate that you are able to undertake and complete a substantial piece of work at a professional level. It is also intended as an opportunity to put into practice various concepts and techniques studied on a Computing degree programme. The project will help the student to acquire a deeper insight into certain aspects of computing, a particular application domain, or both.

If you are undertaking the Project in collaboration with an external organisation, then you may also wish to arrange for a co-supervisor from outside the University. All such arrangements

must be formally approved by the Director of Learning and Teaching for Computer Science.

1.3 Scope and Content of Project

The Project may focus on any of the themes within the degree programme. The student is generally expected to apply existing tools and techniques in order to *solve a well-defined problem*. We stress that the problem itself is at the heart of the Project, and provides *context* for the work that you do. This includes the project objectives, the literature survey, any studies performed, any requirements that you set, the software developed and the tests / analyses performed.

In keeping with the overall purpose, it is desirable that the output from the project should be ‘real’ in the sense that it uses real world data, or meets a real need of some external organisation or class of end-users, or that it could be used to do a real job for someone other than the author. But note that it is not expected to develop a full product that is ready to be released in the marketplace. You are only a one-person team. You can typically only focus on one part of the solution. The development of a project whose outcome embodies little or no individual contribution by the student will not be enough (such as a literature survey or a descriptive evaluation of work that was carried out entirely by others).

It is essential that the Project be carried through to some appropriate conclusion, i.e. a clearly identifiable achievement, which can be shown to be of value to its potential end users. A good example past project was done by a student who took group photos and used AI to recognise faces and change closed eyes to open ones, and unsmiling faces to smiling ones. Here the project had a clear input and output, and it could be given any real world photo. It did not need a user interface, or to be deployed on the web or anything. It was already a challenging problem to use AI techniques to detect faces, detect eyes open or closed, smiling or not, and to alter those attributes. This was an experimental project to tackle a challenging problem and evaluate how well the solution works.

On the other hand, for another type of project, a traditional software development project, this means that the whole of the initial life-cycle must be taken into consideration when establishing the project objectives and workplan:

- Analysis and specification.
- Design, implementation and testing.
- User validation and acceptance.

Your first concern must be to ensure that the project is *feasible*, so that its objectives can be achieved within a fixed timescale and that necessary resources are actually available. Thereafter, you should ensure that the work carried out at each successive stage has been completed to a *suitable professional standard*. This means that the project is carried out with due attention to both quality and fitness for purpose, and within the broader context of professional issues as conveyed in the BCS Code of Conduct such as information security and freedom-to-operate – in the latter case, requiring the demonstration of efforts in avoiding infringing the intellectual property rights of others.

It should be emphasised that a Project is not expected to result in new discoveries as would be the case for postgraduate degrees. However, you are expected to use *originality* in the compilation of your report, showing *insight*, *technical creativity* and *imagination*. The

wholesale copying of material from books is not appropriate. Please refer to Section 1.6.2 and the Student Handbook for further guidance on the University's plagiarism policy.

1.4 Undertaking a Project

Lectures for the project module are scheduled as per the COM3001 timetable available from timetable.surrey.ac.uk.

You should make regular appointments with your supervisor on at least a fortnightly basis and turn up to discuss the progress of your project. It is important to get started on your project early in the first semester and not to waste time, even though the deadline for the completed report seems a long way off. You will also have relatively little coursework from other modules at this stage, so more time is available for the Project.

The Christmas holidays can also be a good opportunity to do some concentrated work on your Project, but you will need to balance this with your revision for the Semester 1 exams which will be in January / February. You should not leave anything major to be done over the Easter break, since if you do run into problems at that time it may be difficult to contact your supervisor and discuss it with them. Ideally, you should already be well into the details of writing your report before the Easter break.

The Professional Project is worth 45 credits, the equivalent of three 15-credit final-year modules. You are therefore expected to undertake approximately 450 hours study, including the preparation of the report. This translates to about 20 hours per week over a period of approximately 23 weeks which are available for your Project, excluding examination periods. The amount of time required should be taken seriously, and if you do not work hard enough in the first semester you can be sure to run into time pressures later.

It is a good idea to write up work that you do as you go along. Otherwise, when you get to the end of the project, you may have forgotten the details of some of your earlier work. It is also a useful way to organise the work that you have done and can show up gaps that need to be filled in before your move on.

1.5 Expectation Management

Departmental guidelines on what to expect from your supervisor are:

- You can expect a minimum of a 30-min meeting every two weeks with your supervisor. (group or individual)
 - Meetings may be scheduled or unscheduled, depending on supervisor preference.
 - You can expect supervisors to reply to emails within 3 days. If you do not receive a reply in this time-frame, please do send a gentle reminder. If you are having difficulty reaching your supervisor, please contact the Undergraduate Projects Coordinator.
 - Whenever you feel the need for a supervision meeting, be clear and specific in your request (e.g. I would like to make an appointment for a 30-min meeting to discuss progress on my final year project, some time in the coming week; please let me know when is suitable.). Otherwise it is easy to get into situations where you think you have requested a meeting, but your supervisor thinks otherwise.
 - Your supervisor does not need to be pro-active in requesting meetings with you. Rather, it is your responsibility to be pro-active in requesting supervision meetings.
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1.6 Legal, Social, Ethical and Professional Issues

One aim of the project is to provide students knowledge about legal, social, ethical and professional (LSEP) issues involved in exploitation of computer technology. In the course of the project, students should have clear awareness of external factors which may affect the work of computer professionals, including computer security, plagiarism, software contracts, Intellectual Property (IP), and use of IPs to protect software, and design, implement and maintenance of trustworthy software.

1.6.1 IP Issues

On admission, undergraduate students are bound by the University Code of Practice in relation to Patenting and Exploitation of Inventions (IP Code). The IP Code requires undergraduate students not to disclose or use without authorisation any confidential information to which they may have access. It also states that each undergraduate student may be required to assign to the University or its nominee any intellectual property rights (whether patentable or otherwise) that he/she is considered to have acquired whilst studying at the University of Surrey in return for a fair proportion of any net receipts in accordance with the terms of the IP Code.

More information is available online at:

<https://www.surrey.ac.uk/sites/default/files/intellectual-property-code.pdf>

Any student concerned about intellectual property rights (IPR) issues should consult the Undergraduate Projects Coordinator. Similarly, the project supervisors will liaise with the Undergraduate Projects Coordinator as necessary in case of IPR issues arising.

1.6.2 Plagiarism

In completing the project, students are required to demonstrate literacy skills, including gathering relevant references and performing critical literature review, citing references properly, and understanding plagiarism and copyright.

The University takes plagiarism very seriously. Further details, including University policy on dealing with cases of alleged plagiarism, can be found in the Department of Computer Science Student Handbook.

1.6.3 Further Information About LSEP Issues

Projects are required to obtain ethical approval. General information about ethics provided by the University can be found at: <https://research.surrey.ac.uk/ethics>. There is an ethics form that can help you determine whether your planned research needs to be reviewed based on UEC's ethical review criteria. When such criteria need to be addressed for your project, you should discuss them with your supervisor and the coordinator of the final year projects.

Furthermore, in the report's ethics section, students are expected to exercise their judgment on how their project meets the requirements of the BCS Code of Conduct that can be found at <http://www.bcs.org/category/6030>. The SoE should include the students' reflections on the *legal*, *ethical*, *social* and *professional* considerations of the research conducted within their project. The basic principles that should be applied should (at least) include:

- Do no harm
- Informed Consent
- Confidentiality of Data
- Social Responsibility (How will the results of your project could potentially be used?)

A detailed presentation of these principles and how they should be addressed within the SoE, will be included in the Lecture on LSEP issues that will take place in the Week 7 lecture.

Do no harm: When conducting research on sensitive subjects (e.g., human subjects or safety-critical pieces of software), try to minimize harms and risks and maximize benefits; respect legality, confidentiality and privacy and take special precautions. In cases where vulnerability analysis and/or penetration testing of sensitive systems is included, the ethical question that arises is whether good or harm will result from the research. Overall, you should demonstrate that you are abiding fully by appropriate laws unless there are strong ethical reasons not to - e.g., you should reflect on, and analyze, how your project aligns with the relevant legislation such as the Computer Misuse Act (CMA).

Informed Consent: In case your project involves the active participation of human subjects, you should consider whether the request for an informed consent is applicable. The participants must be told in advance the purpose of the test, what they will be expected to do and any likely consequences (including side-effects). Such a consent can be considered to balance between an ethical obligation and a legal compulsion depending on the nature of the project. All these issues should be documented in your Statement of Ethics.

Confidentiality of Data: Any information that you collect regarding individuals must be kept confidential and *not* include any identifying information. It is worth noting that there is a legal obligation to protect such data. The legality aspects of data protection balance the *legitimate* need to process personal data and the *protection* of individuals' privacy. In your Statement of Ethics, you should reflect on how your project is aligned with the eight principles of the Data Protection Act (DPA).

Social Responsibility: In general, this reflects the strive for using any research outcomes towards promoting social good and prevent or mitigate social harms. Of course, you don't have almost no control over this - but are there any ways in which the results of your research might be mis-used. For instance, in security-related projects that include vulnerability analysis and penetration testing of systems, could any identified exploits could be used not only by system administrators for enhancing the security profile of the systems but also from malevolent users trying to actually exploit the system? What are the repercussions in this case? All such issues should be addressed in your Statement of Ethics.

1.7 Complaints and Queries

The principle described in the Department of Computer Science Student Handbook applies to your final year project. The first step is to raise your query with the person that has the most knowledge of your progress in your final year project. In the case of the final year project, this will be your project supervisor.

If you consider it inappropriate to contact your project supervisor, or you have concerns about your supervision, then you can raise your concern with the Undergraduate Projects Coordinator. If your project supervisor is also the Undergraduate Projects Coordinator then you should contact, in turn, the Year 3 Coordinator or the Undergraduate Director of Studies.

Chapter 2

Deadlines

2.1 Important Dates

The important dates are all on surrey learn under course materials.

All electronic submissions are through SurreyLearn. Do not hand in deliverables to your supervisor, and do not email deliverables to the Undergraduate Projects Coordinator; all submissions need to be recorded centrally. In general, feedback for the two formative assessments may be expected within three weeks of the relevant submission.

2.2 Units of Assessment

There is only one Unit of Assessment in the Professional Project: the Final Submission. This carries a 100% weighting towards your overall mark for the Professional Project.

You are also expected to demonstrate your implementation to your supervisor near the end of the project. The Implementation Demonstration provides an opportunity to demonstrate your near final implementation to your supervisor, and for the supervisor to check the quality, originality, and depth of your solution. Although the feedback you receive for this component will be formative, the strength of the demonstration may have an impact on your overall grade.

2.3 What if I miss a deliverable deadline?

No marking penalty will apply to an assessed project deliverable submitted by a student within the limits of an agreed extension. If a student is not granted an extension, and hands in an assessed project deliverable late, a marking penalty will be applied in accordance with the University late penalty scheme. Extensions can only be granted following an application for Extenuating Circumstances. Extenuating circumstances applications are processed by the FEPS Extenuating Circumstances Staff (feps-ec@surrey.ac.uk), and not the CS staff.

Note that the Final Submission consists of multiple deliverables. If any of these deliverables is late, the whole unit of assessment will be considered a late submission, even if any other part was submitted on time.

If a student does not submit a Draft Report deliverable, their supervisor will not see the report before it is actually submitted as the Final Report and hence the student will miss the chance to receive comments and feedback on their work before it is submitted as the Final Report deliverable.

CHAPTER 2. DEADLINES

For the interim discussion, once a date and time are agreed with the examiner, this cannot be changed unless Extenuating Circumstances apply.



Chapter 3

Project Selection and Supervisor Allocation

3.1 Discussing your project topic with your supervisor

Once you have been allocated a supervisor, you are expected to discuss suitable project topics with your supervisor. This may include an idea of your own that you would like to pursue further and believe is appropriate for a Professional Project. To do this, it will be helpful to write a brief description of the project you are proposing and an outline of the work that you wish to undertake. Your supervisor will be able to guide you whether the proposed topic is of the appropriate nature for a Professional Project.

If you are stuck for ideas, you should start by discussing with your supervisor the general areas you are interested in. Your supervisor may be able to suggest topics based on this and on their research background. Note that in this case you are still responsible to take ownership of your project and develop it as required.

In thinking about the projects consider the following:

- To get a good mark you should do something reasonably sophisticated. If the final project is entirely based on what you learnt in first year, then you can't expect a very high mark. On the other hand if you use more advanced knowledge, such as the level in 3rd year (or even beyond this), then you can get a higher mark. E.g. advanced AI ideas or security ideas (the two specialisms in the dept.).
- You did a software engineering project in 2nd year. Some students do another software engineering project for their FYP. They talk about the different methodologies, functional, nonfunctional requirements, do UML diagrams, etc. Yes, you can do that, and you can pass. But it's disappointing to see a rerun of what you did in second year. It's more interesting to try something different, e.g. a more in depth focus on a specialist area.
- It is good to take advantage of the expertise of your supervisor. If you work in an area where your supervisor knows a lot, then they can give you a lot of help. You get a head start over someone who has to discover things for themselves.
- Some students have the idea that they need to make some all singing all dancing app, like a product to be released. Especially I am surprised at how many students feel that

whatever they do it also needs to be a web app. This is not necessary. It's often better to put the effort towards a more interesting exploration of a problem. E.g. one nice project I remember was a student who took group photos and used AI to recognise faces and change closed eyes to open ones, and unsmiling faces to smiling ones. It was an interesting problem to investigate, and produced interesting results - absolutely no need to make it into a web app, or any sort of product. Running it from the command line to produce some result files that you can analyse afterwards is completely fine. Of course you can do a web app, and that makes sense for some projects. But don't feel that **every* project *must have* a web app, and a mobile version too.*

3.2 Resources

The Department is able to provide some hardware / software resources to support FYPs. Any students that would like to make a case for purchasing any hardware / software should first discuss the suitability of such resources for their FYP with their supervisor. Following approval from the FYP supervisor, the student should approach the FYP Coordinator, who will make a judgement about any resources purchased on a case-by-case basis. Any resources purchased by the Department are to be considered as a loan, and *must* be returned to the Department upon completion of the Project.

3.3 Undertaking a Collaborative Project

A Collaborative Project is a Professional Project which includes collaboration with an industrial partner, the so-called 'collaborating organisation'. Typically the collaborating organisation will have already identified the final year student for the project it proposes, and will have liaised with the Undergraduate Projects Coordinator in accordance with the guidelines in Chapter 5.

If a student has not been identified, the proposed collaborative project will be advertised to the academic staff. Supervisors will be able to suggest such a project as a topic for a student they are supervising if they feel it is suitable. In this case, the supervisor should liaise with the collaborating organisation involved and the Undergraduate Projects Coordinator in accordance with the guidelines in Chapter 5.

The Department is keen to promote and support professional projects which include collaboration with industrial partners. However, it is important to ensure that all those involved have a clear understanding of the constraints and limitations which apply to student projects, in order to avoid subsequent misunderstandings or unreasonable expectations. If a student chooses a collaborative project then the guidelines in Chapter 5 must be read.

Chapter 4

Project Deliverables

Please submit all deliverables of your Professional Project to the Undergraduate Office and/or via SurreyLearn as specified below. Do not hand in deliverables to your supervisor, and do not email deliverables to the Undergraduate Projects Coordinator; all submissions need to be recorded centrally.

4.1 Project Synopsis

(Due dates on Surrey Learn)

The project synopsis allows you to receive feedback on your written work. You are asked to produce a short (3-page report) describing your progress thus far. A LaTeX template for this report has been provided, which you may choose to use (but are not required to do so).

The project synopsis should contain the following components.

- Main report
 - Introduction to problem, aims / objectives (half page)
 - Literature review / background (1 page)
 - Technical overview (1 page)
 - Workplan, including risks and timeline (half page)
- References (doesn't count towards the page limit) - as many as you need

4.2 Interim Discussion

(Due dates on Surrey Learn)

The interim discussion is an approx 20 minute discussion with you and an interim discussion examiner to discuss your project and progress to date. In brief, the discussion focuses on the project objectives, the outcomes, and your plan of action in order to achieve the objectives in the duration of the project. The specific date will be announced by the examiner. Once a date and time are agreed with the examiner, this cannot be changed unless Extenuating Circumstances apply. Refer to the relevant assessment form for details on how this is assessed.

4.3 Draft Report

(Due dates on Surrey Learn)

You should submit a copy of the current state of your written report and a copy of your planned table of contents, by no later than the prescribed deadline. This is a minimum requirement; your supervisor may agree to review your draft report earlier.

Note that this report does not have to be printed and will not contribute to the final grade of the module. Instead you should submit the report electronically via SurreyLearn. There is an expectation that you submit a draft report because it is important for you to receive feedback on your written report at this stage. You can expect to receive feedback from your supervisor within three weeks.

4.4 Final Submission

(Due dates on Surrey Learn)

The final submission consists of the following deliverables:

1. An electronic copy of the report, submitted on SurreyLearn. This **must** be a single PDF file less than 20 MiB in size. Be particularly careful with the sizes of included images, which can easily cause the PDF to be too big. See also the 50 page limit described in the final section of this handbook (Recommended Format For Report → What is an appropriate format for the report?).
2. A video presentation of no longer than 5 min, containing a demonstration of your software or a presentation of results, as appropriate, submitted on SurreyLearn. This **must** be a single video file less than 500 MiB in size. Uploading large files is best done on campus from a wired connection. The video file **must** be playable with the VLC media player¹. It is recommended (we're not so fussy as long as it plays) that the video is encoded to H.264/MPEG-4 AVC² at VGA resolution (640 × 480) and a bit rate of no more than 1.5 Mbps.
3. An archive containing any software developed as part of the project, submitted on SurreyLearn. This **must** be a single archive file less than 500 MiB in size. Uploading large files is best done on campus from a wired connection. If your project requires the submission of data files that exceed this limit, please discuss this with the Undergraduate Projects Coordinator³

If any of these submissions is late, the whole unit of assessment will be considered a late submission, even if any part was submitted on time. Refer to the relevant assessment form for details of the marking scheme.

The video presentation is assessed together with the report, and should include a presentation of the software that you have created or your research findings, explaining what you have achieved and what you have learnt from the experience. A demonstration of your software

¹This can be downloaded from: <http://www.videolan.org/index.html>

²For a free encoder, go to: <http://www.videolan.org/developers/x264.html>

³The usual solution is to submit the archive on DVD media to the UG office; however, this should only be done on the advice of the Undergraduate Projects Coordinator.

should also be included where appropriate. Refer to the relevant assessment form for details of the marking scheme.

The submitted archive is to be treated as an appendix, and would normally include:

- Source files for programs developed.
- Any Makefiles (or development environment setup)
- Instructions for building any programs from the included source code or executable files. If your program was not built on a departmental computer system then you must say how it was built on whatever system you used. If you have used your own computer, for example a PC, then you must say how to install the files needed to run your code.
- Results files or data sets.
- Other files specific to your project.

You must make sure that the archive contains enough information so that your examiners have the information needed to run any programs which you have produced as part of your project.

4.5 Project Assessment

Final Year Projects will be assessed by two examiners, one of whom may be your supervisor.

A well planned, well executed project of sufficient technical complexity will gain high marks. Projects that have a mediocre, or poor write up, but are sufficiently challenging will gain high marks for technical contribution, but may lose marks in the write up. Likewise, a well-written report that fails to address a problem of sufficient technical difficulty will fail to attract high marks in the technical contribution under Methodology and implementation section. Examples of the latter include simple web-development or data science projects that have not gone beyond instantiation of well-known Python libraries.

The markers are well aware that not all projects are successful, so incomplete results are acceptable. However, it is important to develop your work in stages to manage risk. Identify and develop a minimal viable project that you can expand on over the course of the year.

Gauging the difficulty of a project can be difficult, therefore it is very important for you to discuss your project with your supervisor on a regular basis. The interim discussion will provide another opportunity for formative feedback, so please take this milestone seriously. It is also important for you to keep accurate records of your progress, e.g., using a Git repository, and to set aside enough time for writing. If you are having difficulties at any stage that you cannot resolve with your supervisor, please contact the Undergraduate Projects Coordinator.

Chapter 5

Collaborative Projects

5.1 Overview

The Department is keen to promote and support final year projects which include collaboration with external organisations. Collaboration provides additional purpose and drive to the students' work and offers the opportunity to further develop their professional skills. However, it is important to ensure that all those involved have a clear understanding of the constraints and limitations which apply to student projects, in order to avoid subsequent misunderstandings or unreasonable expectations. This document aims to provide that understanding, of the context in which a student project is carried out.

5.2 Objectives

The final year project is a major component of the final year programme, intended to give the individual student an opportunity to demonstrate that they can apply themselves professionally to a significant piece of work. For any degree accredited by the British Computer Society, the main product of the project must be of an engineering nature which, in its broadest sense, means that it must, potentially, be of use to somebody other than the author.

In contrast, the project is assessed on the basis of a report which is, essentially, a critique of the student's own approach to and execution of the project. This report, whilst it is normally available to the collaborating organisation in the case of collaborative projects, may not be of direct relevance, since it will show how the student has developed his/her understanding and insight, rather than what should be done to develop the product further.

5.3 Timescale

The final year project will run from October until May for the current academic year.

The student effort available for a final year project is approximately **450 hours**. The effort is distributed throughout the academic year alongside the other modules which the students must study. It must be recognised that the generation of the required interim deliverables, which are themselves intended to support the development of the project, is included in these figures. The interim deliverables include a discussion and draft report. Given the academic objectives above, it may be that priority will have to be given to the production of these reports rather than to completion of the product.

5.4 Resources

The Department has hardware/software resources to support projects, and it is often the case that collaborating organisations use specific products not available within the University environment. It is unlikely that the Department will be in a position to acquire the necessary items, and it normally falls, therefore, to the collaborating organisation to make such hardware/software available, often on a loan basis, for the duration of the project.

5.5 Supervision

A collaborative project is normally supervised jointly by a member of the department's staff and a member of the collaborating organisation. The degree of commitment of the latter is flexible; the only request is that a brief statement of the value of the students' work is made available to the examiners.

5.6 Intellectual Property Rights

For major collaborations, the University has a code of practice for the assignment of intellectual property rights and other benefits where the result of the collaboration is likely to lead to an exploitable product. The nature of final year projects, however, is often such that it is to the mutual benefit of all parties concerned if a less formal approach is taken, provided that the interests of all are safeguarded.

5.7 Liability

It is essential that the environment for the project is recognised. The student is pursuing the project for academic reasons, and the primary interest is academic. Neither the student nor the department can be held liable if, for any reason, the project is not completed within the timescale available. There will be no obligation on the student to complete the product following submission of the final report, or to expend additional effort to the detriment of other studies.

Similarly, whilst every effort is made to ensure that students approach their projects in a professional manner, no subsequent liability can be accepted for defects in the product, which is normally made available to the collaborating organisation entirely at the collaborating organisation's risk.

5.8 Setting up a Collaborative Project

Projects can be set up in cases when a student has already been identified, or can be set up in advance of identifying a student. This section covers each of these cases.

5.8.1 Student has already been identified

When a student has already been earmarked for the project then the next step for the collaborating partner is to provide the following information:

1. Describe the aims of the project in a brief paragraph.
2. Name of student who is going to be working on the project.
3. Comment on whether the project is a follow on from the work carried out by the student as part of his/her placement at the collaborating company.
4. Name and job role of the industrial supervisor.
5. Identify relationship of industrial supervisor with student.
6. How often would the industrial supervisor be able to provide supervision?
7. What computing resources would be required to carry out the project? For example, specialised software/hardware.
8. If the resources identified above were not readily available within the Department of Computing would the collaborating organisation be prepared to provide these facilities as necessary throughout the project period? It would be necessary to ensure that the student is not disadvantaged if the resources are recalled at a vital time (maybe including the time that the project report is being written)?

The information should be emailed to the FYP Coordinator. The coordinator will then review the information and discuss the project with the student. The coordinator will also help identify an appropriate academic member of staff to supervise the project within the Department and then the academic supervisor will take the lead in the collaboration from then onwards for the lifetime of the project.

5.8.2 Student has not already been identified

In this case the next step is for the collaborating partner to provide the following information:

1. Describe the aims of the project in a brief paragraph.
2. Name and job role of the industrial supervisor.
3. How often would the industrial supervisor be able to provide supervision?
4. What computing resources would be required to carry out the project? For example, specialised software/hardware.
5. If the resources identified above were not readily available within the Department of Computing would the collaborating organisation be prepared to provide these facilities, as necessary throughout the project period? It would be necessary to ensure that the student is not disadvantaged if the resources are recalled at a vital time (maybe including the time that the project report is being written)?
6. Identify any security clearance requirements for the student working on the project.

The information should be emailed to the FYP Coordinator. The coordinator will then review the information and help identify an appropriate academic member of staff to supervise the project within the Department. The academic supervisor will take the lead in the collaboration from then onwards for the lifetime of the project.

Chapter 6

Exceptional Cases

6.1 Resitting a project

If you have failed your final year project in the previous academic year or have withdrawn from the project, and no Extenuating Circumstances have been agreed, the rules related to module resits apply. Please refer to the Department of Computer Science Student Handbook for further details.

6.2 Extenuating Circumstances

Students may apply for an extended submission deadline due to extenuating circumstances. Extensions can only be granted following University procedures for an application for Extenuating Circumstances. Extenuating circumstances applications are processed by the FEPS Extenuating Circumstances Staff (feps-ec@surrey.ac.uk), and not the CS staff.

Chapter 7

Recommended Format For Report

7.1 What should the report contain?

It should cover the objectives, methods and achievements of the project. It should conclude with a critical appraisal of your own work, and reflections of what you have learned in the process. See the Marking Criteria for details of the assessment criteria for the report. In summary, marks will be awarded to reflect the technical presentation of your work, including English structure, grammar, layout and so on, as well as achievements and argumentation presented in the report. You do need to discuss in the report how you met your objectives and critically evaluate your achievements with respect to the aims and objectives of your project.

You should include code in your report in order to add clarity to the discussion in the text, or you need to show how different versions were produced. This will be different for everyone. For example, it would be sensible to show how a difficult algorithm was coded up but maybe not all the GUI code needs to go in it. You are *not* required to print out your code and include it all in an appendix. You *must*, however, include all your code in the submitted archive.

7.2 What is an appropriate format for the report?

The main body of the report, from first page of introduction up to (but not including) the references, should not exceed 50 pages. Just to be clear, because somebody asked, this 50 pages does not include: the abstract page, acknowledgements page, the list of figures page, the list of tables page, the glossary page, the abbreviations page, the table of contents page, the references page, and the appendix pages. The font size for the main body should not be less than 11 point. The spacing between lines is recommended to be single. You won't automatically fail for exceeding the 50 page limit, but you may be reducing rather than increasing your mark. 50 pages is an upper limit, we are not saying everybody needs 50 pages, we just want to stop students making their document too long. Shorter projects can still get a high mark.

It is ok to use smaller font size for text in charts or tables, but not too small please. It is ok to include as many pages as you like for appendices that you can refer to in the report.

There is no restriction on the software you use to produce your reports and presentations, e.g. whether you should use LaTeX or Word or OpenOffice. You are encouraged to use a word processing software package that makes the report look professionally prepared. The same applies to your presentation slides.

7.3 Using mathematical word processing software

There is a lot of software currently available that supports your project write up. You may find it helpful to save each chapter as a separate file, as files can become unmanageably large.

Make sure you learn how to cross-reference material as well as use Greek symbols, subscripts and equations. You should pick a reference style after discussing it with you supervisor; the standard styles used in the department are Harvard and IEEE. Further information on referencing, including the use of Harvard and IEEE styles, can be found on the Library & Learning Support website.

7.4 Report Checklist

You need to attend to the details in the report.

1. Make sure you have broken the material down into chapters, sections. Each chapter must start on a new page.
2. Make sure you have a caption below each diagram, graph or figure and table to say what they are.
3. Number sections, figures, tables and important equations so that you can cross-reference them.
4. Make sure that the material in each section fits together coherently and that the section titles etc are an accurate description of the content.
5. Make sure you have included *all* required sections, including LSEP (SAGE form).
6. Make sure the spelling is correct.
7. Make sure that you proof read the report.
8. Make sure references are accurate, dates are correct, and so on.
9. Give your report to friends and colleagues for comment and constructive criticism.

7.5 Report Examples

A selection of past Professional Project reports is available on SurreyLearn. These can be used to judge the normal standard of reports.

Appendix A

Marking Forms

Please see the PDF in the same folder here in Surrey Learn.